

TASK-3

Developing Queries with DML single-row functions and operators

AIM:-

To perform the query processing on database for different retrieval results of queries using DML, DDL single-row operations using aggregate, date, string, indent functions, set clauses and operators.

PROCEDURE:-

Create table for employee schema and insert around 10 retail-store-employees data in this relation perform multi row functions.

Note: These queries assume a sample database with an "retail-store-employees" table containing columns like "retailstore-employee-id", "store-id", "retailstore-employee-name", "salary", and "department", "store-phonenum", "employee-phonenum".

Aggregative operators: In addition to simply retrieving data, we often want to perform some computation or summarization. SQL allows the use of arithmetic expressions. We now consider a powerful class of constructs for computing aggregate values such as MIN and SUM.

1. count: COUNT following by a column name returns the count of tuple in that column. If DISTINCT keyword is used then it will return only the count of unique tuple in the column. Otherwise, it will return count of all the tuples (including duplicates) COUNT(*) indicates all the tuples of the column.

Syntax: COUNT (column name)

Example: SELECT COUNT(*) FROM retail-store-employees;
2-SUM: SUM followed by a column name returns the sum of all the values in that column.

Syntax: SUM (column name)

Example: SELECT SUM(salary) FROM retail-store-employees;

3-AVG: AVG followed by a column name returns the average value of that column values.

Syntax: AVG (n1, n2, ...)

Example: SELECT AVG(salary) FROM retail-store-employees;

4-MAX: MAX followed by a column name returns the maximum value of that column.

Syntax: - MAX (column name)

Example: - SELECT - MAX(salary) FROM retail-store-employees;

SQL > select retailstore-employee-name, max(salary) from
retail-store-employees group by retailstore-employee-name;
DEPTNO MAX(salary)

SQL > select retailstore-employee-name, min(salary) from
retail-store-employees group by retailstore-employee-name having
min(salary) > 1000;

5-MIN: MIN followed by column name returns the minimum value of that column.

Syntax: MIN (column name)

Example: SELECT MIN(salary) FROM emp;

SQL > select retailstore-employee-name, min(salary) from
retail-store-employees group by retailstore-employee-name having
min(salary) > 1000;

SQL String Functions

String functions are used to perform an operation on input string and return an output string. Following are the string functions defined in SQL:

1. UPPER()

Query: SELECT UPPER (retailstore_employee_name) FROM retail_store_employees WHERE retailstore_employee_id = 1;

2. LOWER()

Query: SELECT LOWER (retailstore_employee_name) FROM retail_store_employees WHERE retailstore_employee_id = 1;

3. LENGTH()

Query: SELECT LENGTH (retailstore_employee_name) FROM retail_store_employees WHERE retailstore_employee_id = 1;

4. SUBSTR()

Query: SELECT SUBSTR (retailstore_employee_name, 1, 5) FROM retail_store_employees WHERE retailstore_employee_id = 1;

5. CONCAT()

Query: SELECT ~~CONCAT~~ (retailstore_employee_name, ' ', department) FROM retail_store_employees WHERE retailstore_employee_id = 1;

SQL Date and Time Functions

The Date & Time Functions are built-in functions in the SQL. These functions can be used in SQL queries to perform various date and time operations, such as filtering records based on dates, calculating date differences, and formatting dates for display purposes.

For storing a date or a date and time value in a database, MySQL offers the following data types:

DATE	Format: YYYY-MM-DD
DATETIME	Format: YYYY-MM-DD-HH:MM:SS
TIMESTAMP	Format: YYYY-MM-DD HH:MM:SS
YEAR	Format: YYYY or YY

CURRENT_DATE()

Query: SELECT CURRENT_DATE from dual;

CURRENT_TIME()

Query: SELECT CURRENT_TIME() from dual;

ADDDATE (DATE, DAYS)

SQL> SELECT ADDTIME ('2018-08-01 23:59:59.999999', '1 1:1:1.000000');

DAYOFMONTH (date)

SQL> SELECT DAYOFMONTH ('2018-02-15');

DAY OF WEEK (date)

SQL> SELECT DAY OF WEEK ('2018-02-15');

DAY OF YEAR (date)

SQL> SELECT DAY OF YEAR ('2018-02-15');

MONTH (date)

SQL> SELECT MONTH ('2018-08-01');

TIME (expr)

SQL> SELECT TIME ('2018-08-01 11:33:25');

Sysdate:

SQL> SELECT SYSDATE FROM DUAL;

next-day:

SQL> SELECT NEXT-DAY(SYSDATE, 'WED') FROM DUAL;

add-months: last-day:

SQL> SELECT LAST-DAY(SYSDATE) FROM DUAL;

months-between:

SQL> SELECT MONTHS-BETWEEN(SYSDATE, HIRE DATE) FROM EMP;

add-monks:

SQL> SELECT ADD-MONTHS(SYSDATE, 2) FROM DUAL;

Least:

SQL> SELECT LEAST('10-JAN-07', '12-OCT-07') FROM DUAL;

Greatest:

SQL> SELECT GREATEST('10-JAN-07', '12-OCT-07') FROM DUAL;

Trunc:

SQL> SELECT TRUNC(SYSDATE, 'DAY') FROM DUAL;

Round:

SQL> SELECT ROUND(SYSDATE, 'DAY') FROM DUAL;

to-char:

SQL> select to-char(sysdate, 'dd/mm/yy') from dual;

to-date:

SQL> select to-date(sysdate, 'dd/mm/yy') from dual;

CHARACTER FUNCTIONS:

initcap(char): select initcap('hello') from dual;

lower(char): select lower('HELLO') from dual;

upper(char): select upper('hello') from dual;

trim(char, [set]): select trim(' <scit> , 'se') from dual;

rtrim(char, [set]); select rtrim(' <scit> , 'e') from dual;

replace(char, search): select replace('jack and joe', 'j', 'b') from dual;

STRING FUNCTIONS:

Concat: CONCAT returns char1 concatenated with char2. Both char1 and char2 can be any of the datatypes

```
SQL> SELECT CONCAT('ORACLE', ' CORPORATION') FROM DUAL;
```

ORACLE CORPORATION

Lpad: LPAD returns expr1, left-padded to length n characters with the sequence of characters in expr2.

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SQL> SELECT LPAD('ORACLE', 15, '*') FROM DUAL;
```

Rpad: RPAD returns expr1, right-padded to length n characters with expr2, replicated as many times as necessary.

```
SQL> SELECT RPAD('ORACLE', 15, '*') FROM DUAL;
```

Ltrim: Returns a character expression after removing leading blanks

```
SQL> SELECT LTRIM('SSMITHSS', 'S') FROM DUAL;
```

Rtrim: Returns a character string after truncating all trailing blanks

```
SQL> SELECT RTRIM('SSMITHSS', 'S') FROM DUAL;
```

Lower: Returns a character expression after converting uppercase character data to lower case.

```
SQL> SELECT LOWER('DBMS') FROM DUAL;
```

Upper: Returns a character expression with lowercase character data converted to uppercase

```
SQL> SELECT UPPER('dbms') FROM DUAL;
```

Length: Returns the number of characters, rather than the number of bytes, of the given string expression excluding trailing blanks

SQL > SELECT LENGTH('DATABASE') FROM DUAL;

Substr: Returns part of a character, binary, text, or image expression

SQL > SELECT SUBSTR('ABCDEFGHID', 3, 4) FROM DUAL;

SELECT SUBSTR('retailstore-employee-name', 1, 4) FROM retailstore-employees;

Instr: The INSTR function searches string for substring. The function returns an integer indicating the position of the character in string that is the first character of this occurrence.

SQL > SELECT INSTR('CORPORATE FLOOR', 'OR', 3, 2) FROM DUAL;

Single-Row Operators:

1. IS NULL

Query: SELECT * FROM retailstore-employees WHERE salary IS NULL;

2. IS NOT NULL

Query: SELECT * FROM retailstore-employees WHERE salary IS NOT NULL;

3. LIKE

Query: SELECT * FROM retailstore-employees WHERE retailstore-employee-name LIKE '%John%';

4. NOT LIKE

Query: SELECT * FROM retailstore-employees WHERE retailstore-employee-name NOT LIKE '%John%';

5. BETWEEN

Query: SELECT * FROM retailstore-employees WHERE salary BETWEEN 50000 AND 100000;

VEL TECH	
EX NO.	8
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	4
RECORD (5)	5
TOTAL (20)	19
DATE	07/08/2020

RESULT: Thus, the developing queries and operators

SQL> select * from retail_store_employees where retail_store_employee_name like '%koteswarao%';

RETAIL_STORE_EMPLOYEE_ID STORE_ID RETAIL_STORE_EM SALARY DEPARTMENT

STORE_PHONENO EMPLOYEE_PHONENO

28129	1569	koteswarao	50000	manager
9515221655	9876543210			

SQL> select * from retail_store_employees where retail_store_employee_name not like '%koteswarao%';

RETAIL_STORE_EMPLOYEE_ID STORE_ID RETAIL_STORE_EM SALARY DEPARTMENT

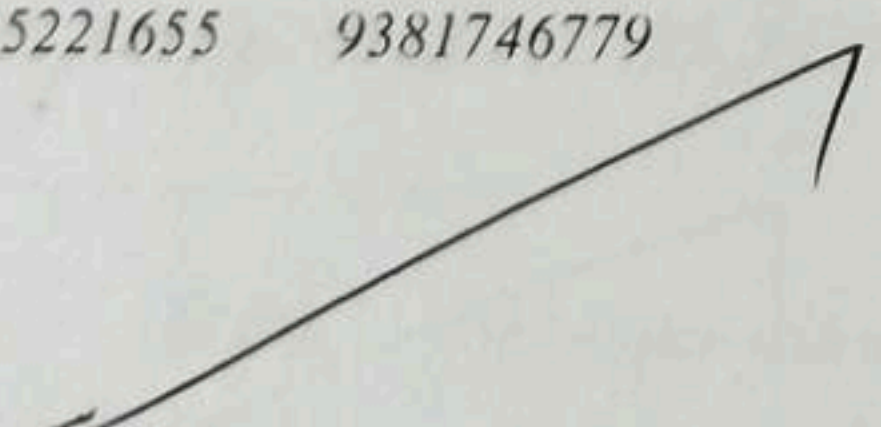
STORE_PHONENO EMPLOYEE_PHONENO

28578	1569	chandrajith	40000	superwiser
9515221655	9866671022			

27830	1569	nithish	60000	hr
9515221655	9963892694			

27578	1569	mahalakshmi	35000	salesman
9515221655	9381746779			

SQL> select * from retail_store_employees where salary between 30000 and 70000;



SQL> select * from retail_store_employees where salary is null;

no rows selected

SQL> select * from retail_store_employees where salary is not null;

RETAIL_STORE_EMPLOYEE_ID	STORE_ID	RETAIL_STORE_EM	SALARY	DEPARTMENT
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STORE_PHONENO	EMPLOYEE_PHONENO
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28129	1569	koteswarao	50000	manager
9515221655	9876543210			

28578	1569	chandrajith	40000	superwiser
9515221655	9866671022			

27830	1569	nithish	60000	hr
9515221655	9963892694			

RETAIL_STORE_EMPLOYEE_ID	STORE_ID	RETAIL_STORE_EM	SALARY	DEPARTMENT
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STORE_PHONENO	EMPLOYEE_PHONENO
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27578	1569	mahalakshmi	35000	salesman
9515221655	9381746779			

SQL> select * from retail_store_employees;

RETAIL_STORE_EMPLOYEE_ID STORE_ID RETAIL_STORE_EM SALARY DEPARTMENT

STORE_PHONENO EMPLOYEE_PHONENO

28129 1569 koteswarao 50000 manager

9515221655 9876543210

28578 1569 chandrajith 40000 superwiser

9515221655 9866671022

27830 1569 nithish 60000 hr

9515221655 9963892694

RETAIL_STORE_EMPLOYEE_ID STORE_ID RETAIL_STORE_EM SALARY DEPARTMENT

STORE_PHONENO EMPLOYEE_PHONENO

27578 1569 mahalakshmi 35000 salesman

9515221655 9381746779

SQL> select count(*) from retail_store_employees;

COUNT(*)

4

SQL> select sum(salary)from retail_store_employees;

SUM(SALARY)

185000

SQL> select avg(salary)from retail_store_employees;

AVG(SALARY)

46250

SQL> select max(salary)from retail_store_employees;

MAX(SALARY)

60000

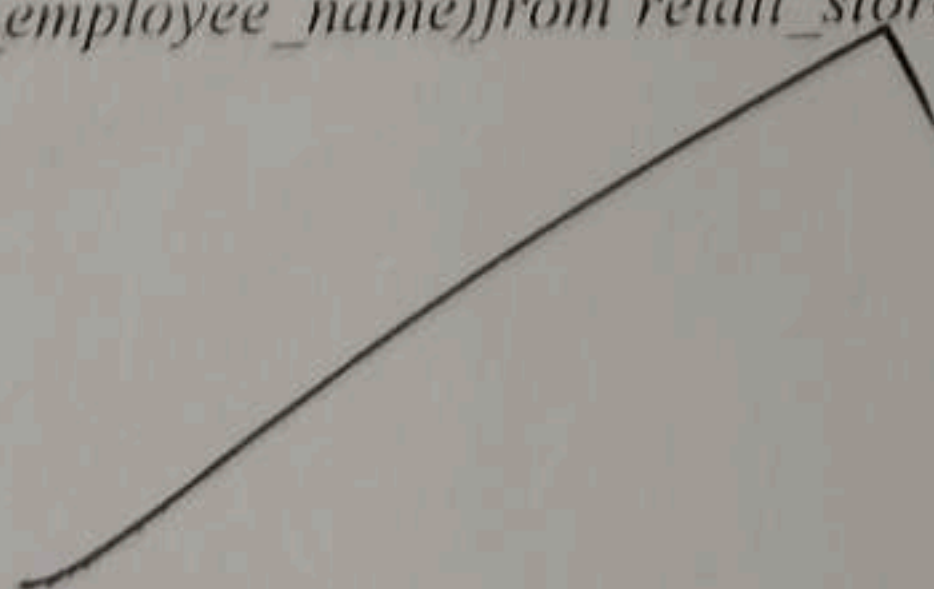
SQL> select min(salary)from retail_store_employees;

MIN(SALARY)

35000

SQL> select upper(retail_store_employee_name)from retail_store_employees where retail_store_employee_id=27830;

UPPER(RETAIL_ST



NITHISH

SQL> select lower(retail_store_employee_name) from retail_store_employees where retail_store_employee_id=27830;

LOWER(RETAIL_ST

nithish

SQL> select length(retail_store_employee_name) from retail_store_employees where retail_store_employee_id=27830;

LENGTH(RETAIL_STORE_EMPLOYEE_NAME)

7

SQL> select substr(retail_store_employee_name,1,4) from retail_store_employees where
retail_store_employee_id=28129;

SUBSTR(RETAIL_ST

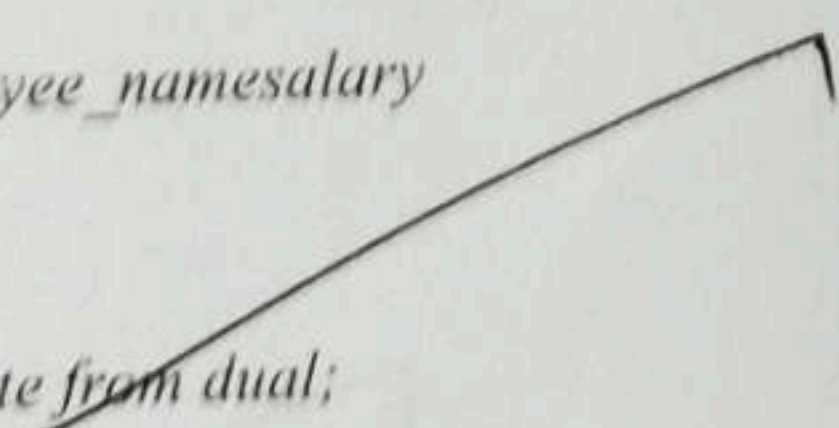
kote

SQL> select concat(' retail_store_employee_name','salary') from retail_store_employees where
retail_store_employee_id=27578;

CONCAT('RETAIL_STORE_EMPLOYEE_NA

retail_store_employee_namesalary

SQL> select sysdate from dual;



SYSDATE

27-AUG-25

SQL> select next_day(sysdate,'wed')from dual;

NEXT_DAY(

03-SEP-25

SQL> select add_months(sysdate,2)from dual;

ADD_MONTH

27-OCT-25

SQL> select last_day(sysdate)from dual;

LAST_DAY(

31-AUG-25

SQL> select least('10-jan-07','12-oct-07')from dual;

LEAST('10

10-jan-07

SQL> select greatest('10-jan-07','12-oct-07')from dual;

GREATEST(

12-oct-07

SQL> select trunc(sysdate,'day')from dual;

TRUNC(SYS

24-AUG-25

SQL> select round(sysdate,'day')from dual;

ROUND(SYS

31-AUG-25

SQL> select to_char(sysdate,'dd\mm\yy')from dual;

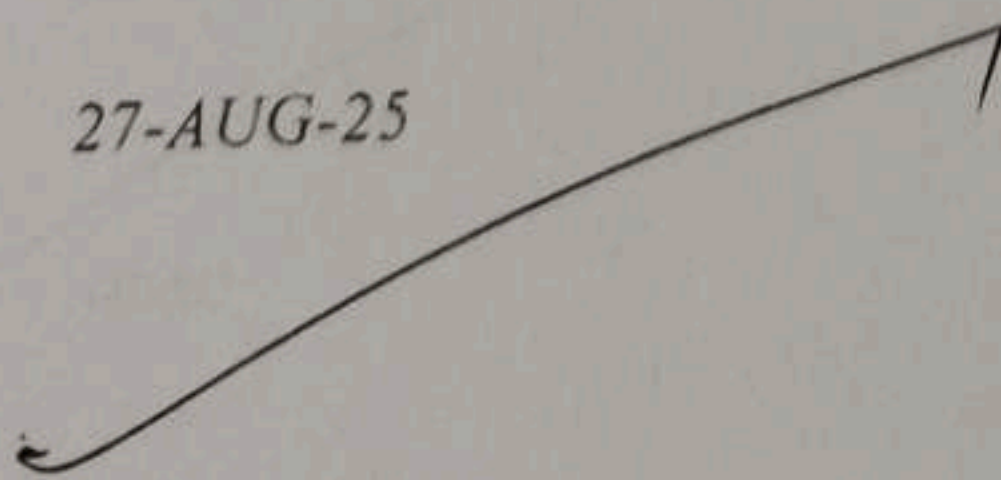
TO_CHAR(

27\08\25

SQL> select to_date(sysdate,'dd\mm\yy')from dual;

TO_DATE(S

27-AUG-25



SQL> select concat('oracle','corpuration')from dual;

CONCAT('ORACLE','

oraclecorpuration

SQL> select lpad('oracle',15,'*')from dual;

LPAD('ORACLE',1

*****oracle

SQL> select rpad('oracle',15,'*')from dual;

RPAD('ORACLE',1

oracle*****

SQL> select ltrim('ssmithss','s')from dual;

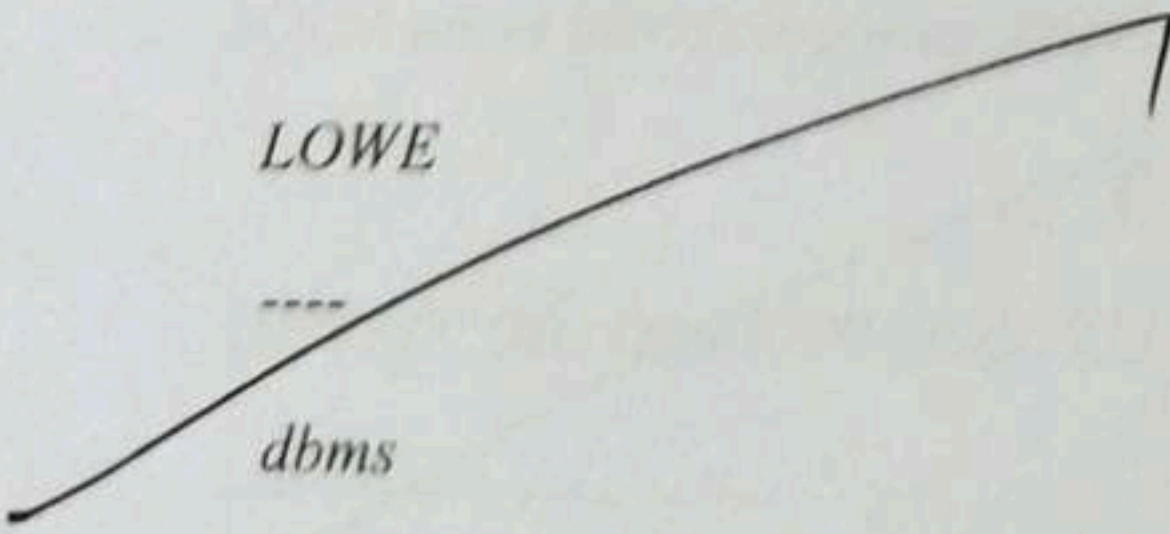
LTRIM(

mithss

SQL> select lower('dbms')from dual;

LOWE

dbms



SQL> select upper('dbms')from dual;

UPPE

DBMS

SQL> select rtrim('ssmithss','s')from dual;

RTRIM(

ssmith

SQL> select length('database')from dual;

LENGTH('DATABASE')

8

SQL> select substr('abcdefghij',3,4)from dual;

SUBS

cdef

SQL> select instr('corporate floor','or',3,2)from dual;

INSTR('CORPORATEFLOOR','OR',3,2)

RE

TAD

28578 1569 chandrajith 40000 superwiser
9515221655 9866671022

27830 1569 nithish 60000 hr
9515221655 9963892694

RETAIL_STORE_EMPLOYEE_ID STORE_ID RETAIL_STORE_EM SALARY DEPARTMENT

STORE_PHONENO EMPLOYEE_PHONENO

27578 1569 mahalakshmi 35000 salesman
9515221655 9381746779

2/10
9/10

13/10

LE8

d